

Business Requirements Document (BRD) - University Library Management System

Version: 1.0
Role: Business Analyst (Entry Level)

1. Executive Summary

The University Library Management System is a digital solution designed to automate student borrowing activities. The system aims to provide real-time visibility into book availability, streamline the reservation process, and manage loan renewals based on specific business rules.

2. Business Scenario

The library requires a system that manages high-volume transactions while enforcing borrowing limits. The system must act as a real-time interface between the student and the library's central database to prevent double bookings and manage overdue risks.

3. Functional Requirements

ID	Requirement Description	Priority
FR-01	User Authentication: Students must log in using a valid Student ID and Password. The system shall display an error for failed attempts.	Critical
FR-02	Online Catalog Search: Users shall be able to search for books and view availability, author, and shelf location.	High

ID	Requirement Description	Priority
FR-03	Reservation Management: The system must validate that the user has not exceeded the maximum reservation limit before confirming a book hold.	High
FR-04	Conditional Renewal: The system shall allow book renewals only if there are no active reservations for that specific book by other students.	Critical
FR-05	Real-time Database Sync: The system must update the central database immediately after every search, reservation, or renewal action.	Critical

4. User Roles (Actors)

- **Student:** Searches for books, places reservations, and manages renewals.
- **Library Database:** Stores real-time status of books and user borrowing history.

5. Business Rules & Logic

Renewal Logic: A renewal request is successful ONLY IF (Book Status = Borrowed) AND (Reservation Count = 0). If a reservation exists, the student must return the book by the original due date.

Reservation Logic: A reservation is successful ONLY IF (Book Status = Available) AND (User Current Reservations < Max Limit).